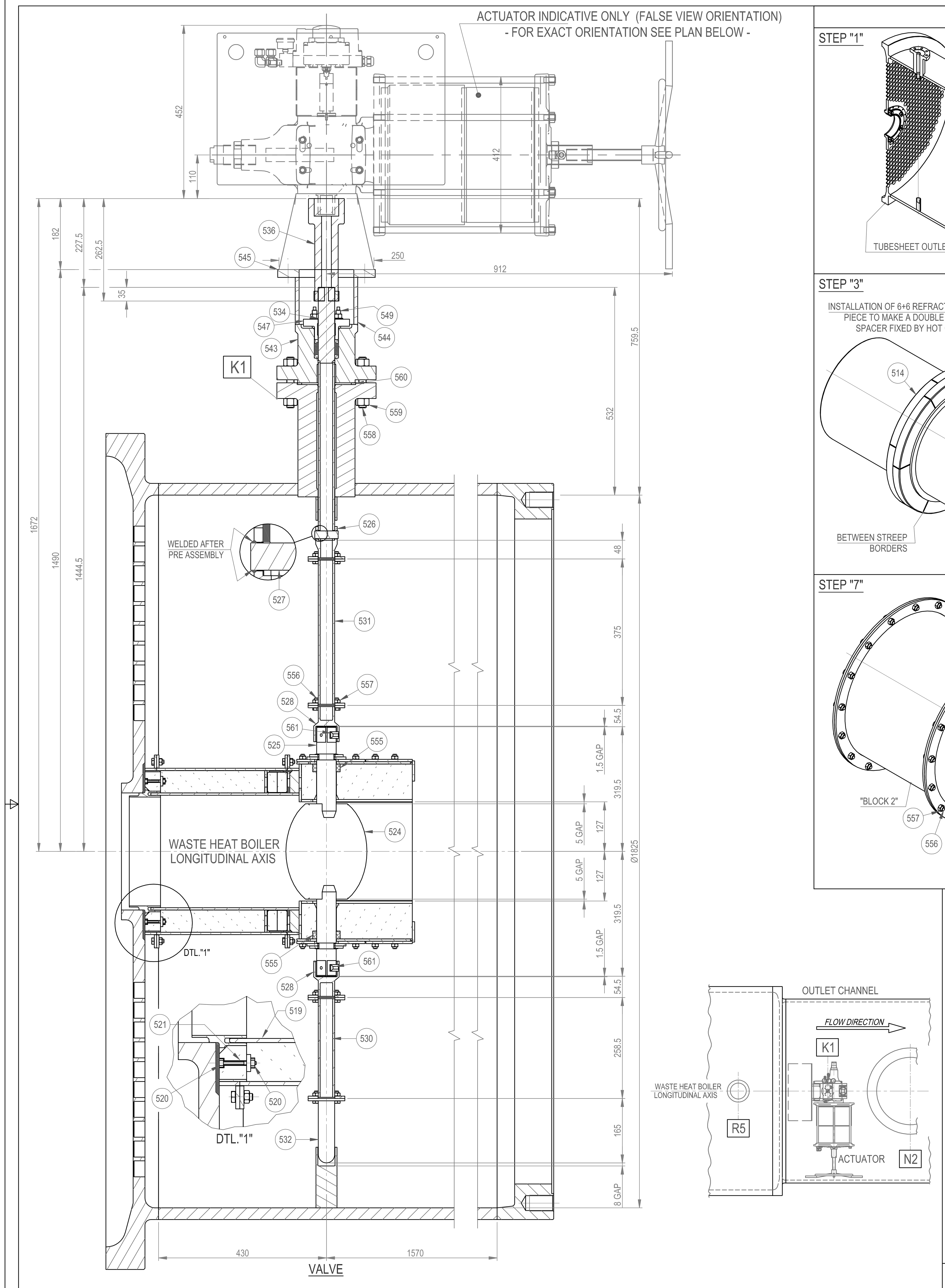
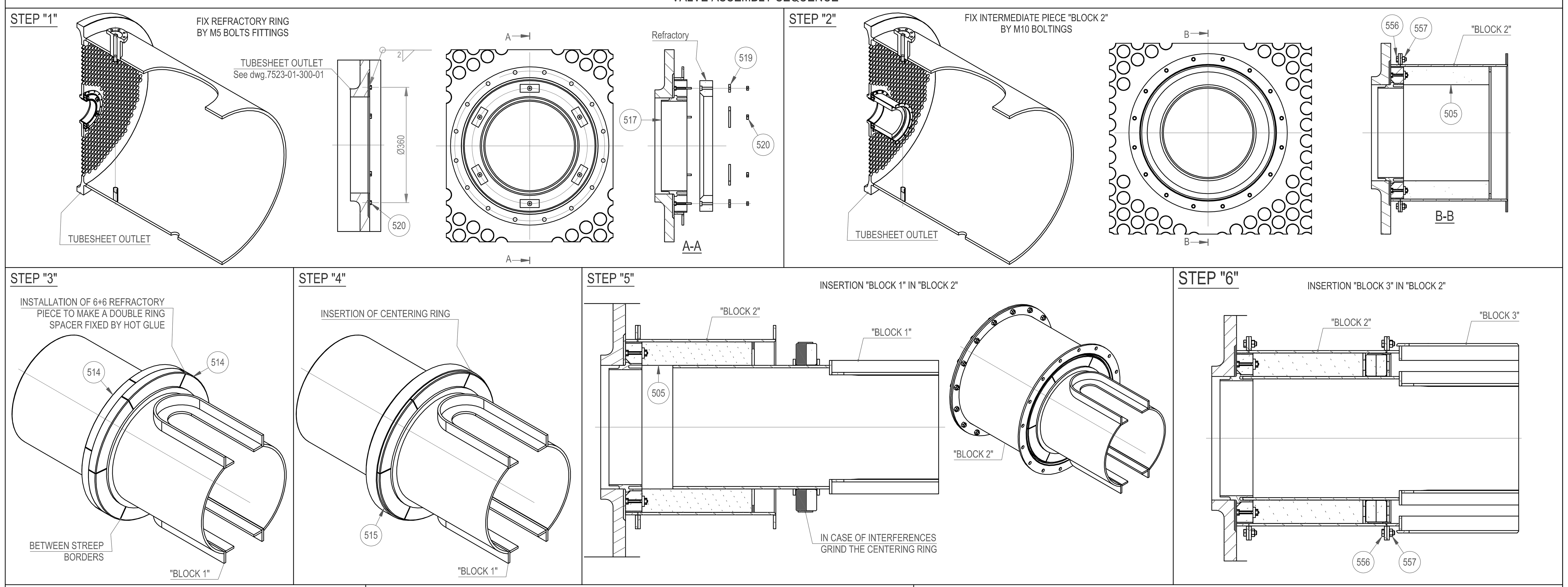
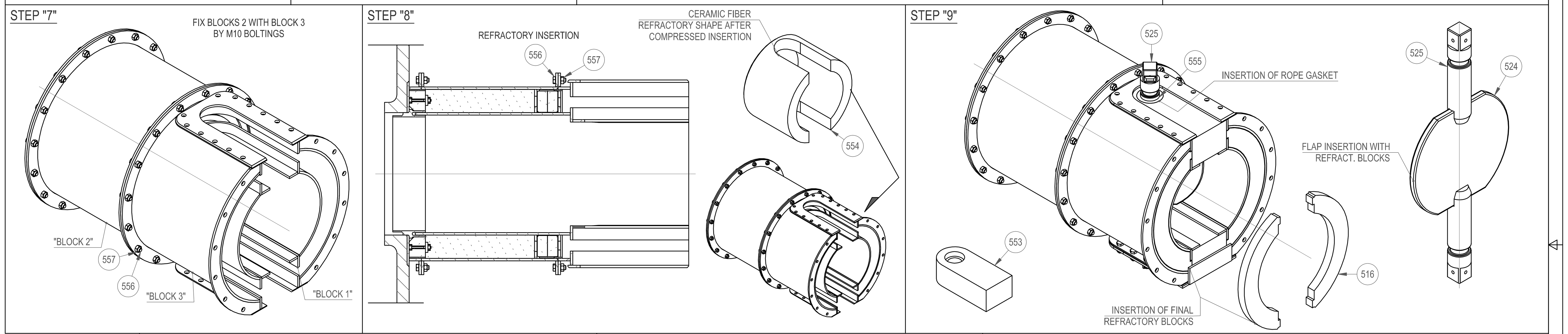
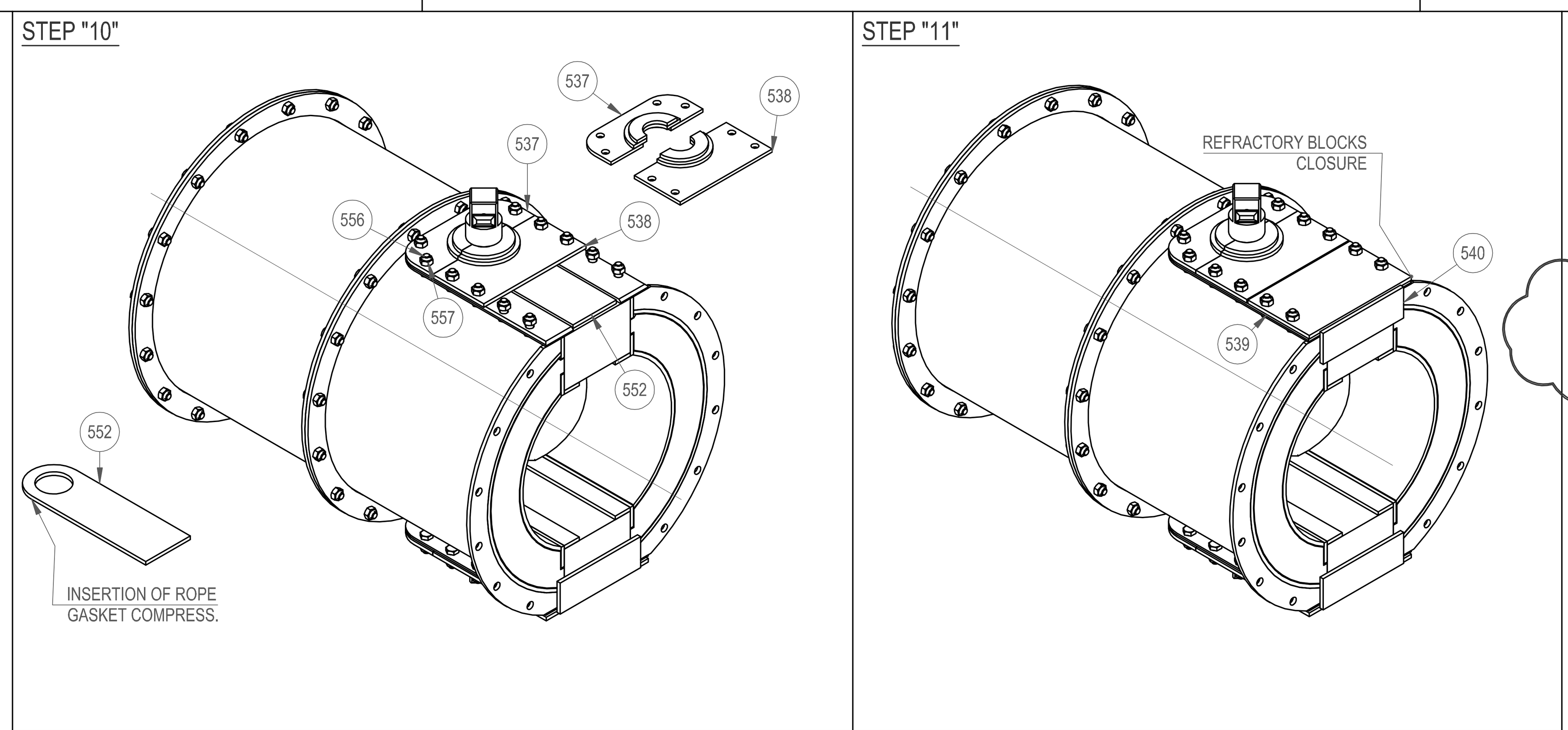
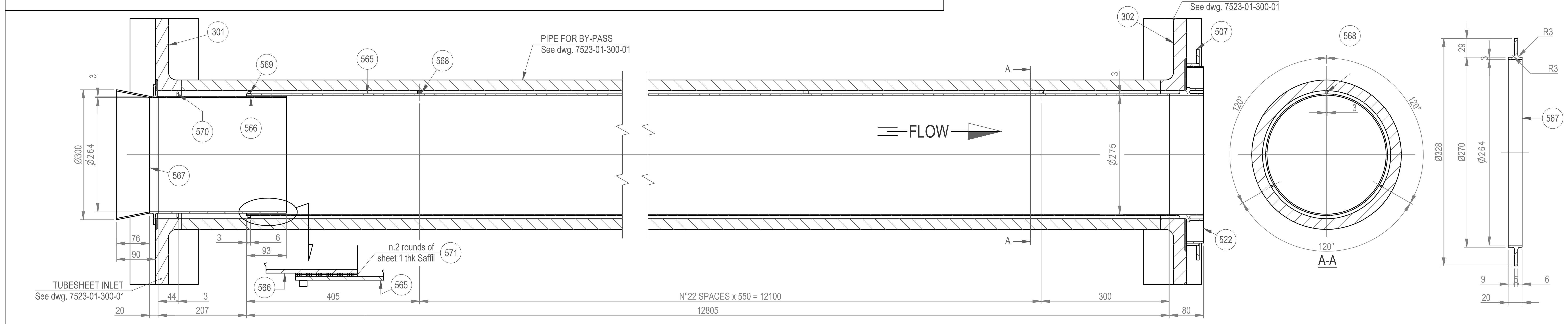


VALVE ASSEMBLY SEQUENCE









POS.	DESCRIZIONE/DESCRIPTION	DIMENSIONI/DIMENSIONS	MATERIALI/MATERIAL	Q.TY	CERTIFICATO CERTIFIED	SPARE PARTS	SPARE FOR TWO YEARS	NOTES
571	Sheet / Carta	#93 x Thk.1 Sv.~1750	Saffil	1				
570	Anello/Ring	Ø294/270 x Thk.3	SB-168 UNS N06025	1	3.1			
569	Anello finale/End ring	Ø295/282 x Thk.6	SB-168 UNS N06025	1	3.1			
568	Piatto/Plate	9.5 x 10 x Thk.3	SB-168 UNS N06025	69	3.1			
567	Anello/Ring	See dwg.	SB-168 UNS N06025	1	3.1			
566	Tubo da lamiera/Pipe from plate	Øe270 x Thk.3 Lg.300	SB-168 UNS N06025	1	3.1			
565	Tubo da lamiera/Pipe from plate	Øe281 x Thk.3 Lg.12805	SB-168 UNS N06025	1	3.1			

MATERIALS TEST SYMBOL (Certification Acc. TO EN-10204) Simbolismo collaudo materiali

- 3.2 Materiale collaudato da ente ufficiale. Material tested by official authority.
- 3.1 Materiale con certificato del produttore. Material with manufacturer certificate.
- 2.2 Materiale senza certificato del produttore. Material without manufacturer certificate.

4
For ITMES numbers below 565 see drawing 7523-01-500-02
Quantity indicated on material list are for one shell.

LISTA MATERIALI / MATERIALS LIST

REV.	DATE	DESCRIPTION	PREPARED	CHECKED	APPROVED
3	22/01/2015	Revised where marked & issued AS BUILT	Picenni	Sarti	Sarti
2	19/11/2014	Revised where marked & issued AS BUILT	Scotti	Picenni	Sarti
1	05/03/2014	General revised	Picenni	Sarti	Sarti
4	25/09/2015	Revised as per client comments	Nani	Sarti	Sarti

Temperature control in the range 355°C - 380°C for all Load Cases between 50% and 100%, for all fouling conditions
EOR = End Of Run
SOR = Start Of Run

Load Case Temperature after mix	Conditions Bypass flow	Temperature out of tubes
100 % 355 °C	EOR leakage only	355 °C
110 % 380 °C	SOR 7.5 %	331 °C
100 % 355 °C	EOR leakage only	355 °C
100 % 380 °C	SOR 8 %	330 °C
50 % 380 °C	EOR 7.5 %	331 °C
50 % 380 °C	SOR 9 %	326 °C

Flap opening versus Load Case

Load Case	Conditions	Valve position (0°=fully open)
110 %	EOR	closed
110 %	SOR	25°
100 %	EOR	closed
100 %	SOR	23°
50 %	EOR	27.5°
50 %	SOR	20°

ALFA LAVAL
SUISIO - BERGAMO - ITALY

CLIENTE / CUSTOMER } Fimco
ORDINE CLIENTE / PURCHASE ORDER } FZMC-13450-R1
IMPIANTO E LOC. / LOCATION & UNIT } Ammonia Urea Project
SERVIZIO / SERVICES } Sec. Reformer Effluent Waste Heat Boiler
TITOLO / TITLE } Internal By-Pass and valve

SCALA/FORMATO SCALE/FORMAT: NO / A1	TIPO / TEMA / TYPE: BXM SPEC.	NUMERO APPAR. NUMBER REQUIRED: 1	SIGLA / ITEM No.: 3-E-1401		
DATA DATE: 29/04/2013					
DISEGNATO DRAWN.: Mazzoleni	DIMENSIONI / SIZE: 13000 x 2025	NUM. DI FABBRICA SERIAL NUMBER: 12542	ANNO DI COS. YEAR OF CONS.: 2013	COMM. / JOB: 7523	NUM. DIS. / DWG. NUM.: 7523-01-500-01 — REV. 4
CONTROLL. CHECKED: Sarti					
APPROVATO APPROVED: Sarti					

DISEGNO DEFINITIVO
Il cliente ha gia preso visione del disegno ed ha approvato le parti non evidenziate dall'ultima rev.

VALID FOR CONSTRUCTION
Drawing already approved by client except where indicated with last revision mark.
DATA: 22/01/2015 FIRMA: Sarti S.

AS BUILT

REV.	ISSUE DATE	DESCRIPTION	DWN. BY	CHKD. BY	APPR. BY	APPR. BY (PROJ.)
4	25/09/15	REVISED AS PER CLIENT COMMENTS	NL	SS	SS	
3	22/01/15	REVISED WHERE MARKED & ISSUED AS BUILT	SP	SS	SS	
2	19/11/14	REVISED WHERE MARKED & ISSUED AS BUILT	GPS	SP	SS	
1	05/03/14	GENERAL REVISED	SP	SS	SS	
0	29/04/13	ISSUED FOR APPROVAL	PM	FG	SS	

OWNER: LORDEGAN PETROCHEMICAL COMPANY
Project: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR: HAMPA ENGINEERING CORPORATION
DETAIL ENGINEER: HEIDCO

OWNER: LORDEGAN PETROCHEMICAL COMPANY
PROJECT TITLE: LORDEGAN AMMONIA AND UREA PROJECT
CONTRACTOR: HAMPA ENGINEERING CORPORATION
P.O. NO.: 2103-POR-A-HM-255
REQ NO.: 2103-REQ-A-HM-255
DOC. NO.: 2103-A-VD-255-A1-MD-027

TITLE: INTERNAL BY-PASS AND VALVE
Project No: 2103 — Doc. No. 7523-01-500-01 / 2103-A-VD-255-A1-MD-027 — REV. 4

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